

Week 8 Project Guide

Forecasting: Classical to Modern

[Role Requirements](#) · [Notebook Walkthrough](#) · [Submission Guide](#)

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How to use this document: Read **Your Role** first — every modelling decision in the notebook flows from the business framing. Use the **Notebook Walkthrough** as a per-part checklist. Check the **Submission Requirements** carefully before uploading — a missing artefact or an empty Reflect cell will fail the submission. All links are clickable.

1 YOUR ROLE

The business scenario you are operating in for this entire assignment

You are a **Quantitative Analyst at an investment research firm**. The portfolio management team currently sets 12-month equity exposure targets using a single rule: *last month's index level is the best estimate of next month's level* — the random-walk (naïve) baseline. Your Chief Investment Officer asks two questions: (1) “Can any quantitative model beat the random-walk baseline reliably over a full 12-month horizon — even accounting for the Efficient Market Hypothesis?” and (2) “If you had to commit one model to production for monthly equity allocation,

which would you choose — and can you defend that choice statistically, not just with a lower MASE?”

Your task: build and compare six forecasting models on ~360 months of S&P 500 data (1990–2019), evaluate on a 60-month held-out test window (2020–2024, including the COVID crash), and produce a 200-word investment memo that answers both questions above.

Responsibility	What it means in practice
Prepare the data	Part 1 (Q1–Q5): download via yfinance, build DatetimeIndex, resample, simulate missing data, run ADF + KPSS stationarity tests, plot ACF/PACF.
Build classical + SARIMA	Part 2 (Q6–Q10): naïve baseline (MASE = 1.0), Holt-Winters, SARIMA Box-Jenkins workflow (identify → fit → Ljung-Box → forecast + CI). All models fit on log-prices; inverse-transform with np.exp().
Build automated models	Part 3 (Q11–Q12): Prophet with changepoint_prior_scale=0.5, cross-validation. Connect Prophet decomposition to W6 composite GP kernel.
Build ML + DL models	Part 4 (Q13–Q16): make_features() with 10 columns, LinearRegression + XGBoost walk-forward CV, LightGBM recursive forecast, quantile intervals. Part 5 (Q17–Q18): MLP and Keras LSTM on 360-month training set.
Evaluate + recommend	Part 6 (Q19–Q22): full 7-metric comparison, error by month/horizon, simple ensemble, Diebold-Mariano test, pickle save. Part 7 (Q23): 200-word investment memo to CIO citing MASE, DM p-value, EMH implications, and 3+ non-accuracy criteria.

Key challenge: The Efficient Market Hypothesis predicts that S&P 500 log-returns are near-white-noise. This means MASE ≈ 1.0 for all models may be the expected outcome — not a failure. MASE < 1.0 is necessary but not sufficient. The CIO wants statistical evidence the improvement is real (DM p < 0.05), not just a difference that disappears due to the COVID structural break. A model with MASE = 0.88 and DM p = 0.22 is not a deployable answer. Your recommendation must address both accuracy and EMH-context significance.

2 DATASET

Monthly S&P; 500 Index 1990–2024 — downloaded via yfinance (run once at session start)

Property	Value	Notes
Source	yfinance (ticker: ^GSPC)	pip install yfinance. Synthetic fallback included if download fails.
Period	Jan 1990 – Dec 2024	~420 monthly observations · auto_adjust=True.
Training split	Jan 1990 – Dec 2019	~360 months. Use TRAIN_END = '2019-12' throughout.
Test split	Jan 2020 – Dec 2024	~60 months — includes COVID crash (Mar 2020: –34%). No model can predict a pandemic shock; this is pedagogically intentional.

Target	S&P 500 Close (USD)	Strong upward trend, weak seasonal pattern, approximate random-walk in returns. Log-transform before ARIMA; log-returns $\approx$ stationary.
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Dataset loading:

```
import yfinance as yf, pandas as pd, numpy as np

raw = yf.download('^GSPC', start='1990-01-01', end='2024-12-31',
interval='1mo', auto_adjust=True, progress=False)

series = raw['Close'].rename('SP500')

series.index = pd.to_datetime(series.index).tz_localize(None)
series.index.freq = 'MS'

TRAIN_END = '2019-12'
TEST_START = '2020-01'

train = series[:TRAIN_END] # ~360 months
test = series[TEST_START:] # ~60 months (includes COVID)

log_train = np.log(train)
```

Why S&P 500? Financial forecasting is the most demanding real-world test for time series models. Stock prices approximate a random walk (Efficient Market Hypothesis), so any model with $MASE < 1.0$ is a genuine finding, not a foregone conclusion. The 360-month training set is large enough for LSTM to learn seasonal structure (unlike the 96-month airline dataset). The COVID test period provides a stress test for structural break robustness — a critical skill in quant finance.

3 NOTEBOOK WALKTHROUGH

Part-by-part guide for completing the assignment

Part 1 — Data Preparation (Q1–Q5)

Q1 (DatetimeIndex)

Download S&P 500 via yfinance. Set `freq='MS'`. Print `index.dtype`, `index.freq`, `len(series)`. Confirm SELF-CHECK assertions pass.

Q2 (Resampling)

Resample to quarterly using `.resample('QE').last()`. Identify decade with highest mean quarterly return; year with largest quarterly crash.

Q3 (Missing data)

Insert 5 NaN values. Compare `ffill`, `bfill`, and `interpolate(method='time')`. Explain the best strategy for monthly financial prices.

Q4 (Stationarity)

Compute `log_series` and `log_returns`. ADF + KPSS on both. Are prices stationary? Returns? Explain ADF/KPSS conflict if present.

Q5 (ACF / PACF)

Plot ACF and PACF of log_returns (36 lags). State whether significant lags exist. What does near-white-noise suggest about EMH?

Part 2 — Classical Models + SARIMA (Q6–Q10)

Q6 (Baselines)

Naive and seasonal naive forecasts. Assert $\text{abs}(\text{naive_mase} - 1.0) < 0.02$. This is the random-walk benchmark — all models are measured against it.

Q7 (Holt-Winters)

ExponentialSmoothing(trend='add', seasonal='add', seasonal_periods=12) on log_train. Report α , β , γ . np.exp() before metrics. Assert hw_mase < 1.0.

Q8 (SARIMA identification)

ACF/PACF of diff(log_train). Propose SARIMA(p,d,q)(P,D,Q,12) with justification. Fit SARIMAX. Report AIC, BIC.

Q9 (SARIMA diagnostics)

Ljung-Box at lags [6, 12, 24]. All $p > 0.05$? QQ-plot of residuals. Adjust order if any lag fails.

Q10 (SARIMA forecast)

get_forecast(steps=H). np.exp() mean and CI. Plot with 95% CI; mark COVID crash. Store in results['SARIMA'].

Part 3 — Prophet (Q11–Q12)

Q11 (Fit Prophet)

Prophet(seasonality_mode='additive', changepoint_prior_scale=0.5) on log_train. Forecast H steps. Plot components. np.exp() before metrics. Why changepoint_prior_scale=0.5 for financial data?

Q12 (Cross-validation)

cross_validation(horizon='365 days', period='180 days', initial='3650 days'). Report mean RMSE. Connect to W6 GP kernel.

Part 4 — Feature Engineering + ML (Q13–Q16)

Q13 (make_features)

Implement with lag_1/2/3/6/12, roll_12_mean/std, month, year, trend. Assert X_train.shape[1] == 10. Report rows dropped by dropna.

Q14 (Linear Reg + XGBoost, walk-forward CV)

TimeSeriesSplit(n_splits=4, test_size=12). Report per-fold log-RMSE. Which fold had highest RMSE? What happened in that window?

Q15 (LightGBM recursive forecast)

LGBMRegressor. 48-step recursive loop with history buffer append. Assert lgb_mase < 1.0. Store in results['LightGBM'].

Q16 (Quantile intervals)

objective='quantile', alpha=0.025/0.975. Report empirical coverage. If coverage < 95%, identify probable cause.

Part 5 — Deep Learning (Q17–Q18)

Q17 (MLP)

MLPRegressor(hidden_layer_sizes=(128, 64, 32)) with StandardScaler. Recursive forecast. Does it beat Linear Regression?

Q18 (LSTM)

Keras LSTM(64), LOOKBACK=12, MinMaxScaler. Recursive 60-step forecast. With 360 training months, does LSTM outperform SARIMA? Discuss.

Part 6 — Evaluation & Comparison (Q19–Q22)

Q19 (Full metrics)

MAE, RMSE, MAPE, sMAPE, WMAE, WMAPE, MASE for all models, sorted by MASE. Which model ranks differently on WMAPE vs MAPE?

Q20 (Error by calendar month)

Mean error by month for best model. Which month shows highest negative bias? Does March 2020 distort the monthly aggregates?

Q21 (Error by horizon)

MAE at each $h=1..H$ for two best models. At which horizon does accuracy collapse? Plot both on the same axes.

Q22 (Ensemble + DM test + pickle)

Simple ensemble of 4 best models. DM test: best single vs ensemble. `pickle.dump(sarima, 'sp500_sarima_v1.pkl')`. Verify it loads.

Part 7 — Investment Recommendation (Q23)

Q23 (Investment memo)

200-word memo to CIO: chosen model (cite MASE), DM significance (p-value), EMH implications, 3+ non-accuracy criteria (interpretability, regulatory explainability, refit latency). Use the template in the assignment notebook.

4 SUBMISSION REQUIREMENTS

Exactly what to submit, how, and what will be checked

4.1 Files to Submit

File	Description	Required?
W8_Forecasting_Assignment.ipynb	Your completed assignment notebook. Fully executed — all 23 questions answered, all plots visible, all SELF-CHECK cells pass, all  Reflect cells filled in. Run top-to-bottom with no errors before submitting.	✓ Required
sp500_sarima_v1.pkl	The fitted SARIMAX result object saved using <code>pickle.dump()</code> . Must reload with <code>pickle.load()</code> and produce a 12-step forecast. This is the deployable artefact — analogous to the .pkl from W5 trees.	✓ Required

Investment Memo (inline in Q23 or PDF, ≤ 1 page)	200-word memo answering both CIO questions: (1) Does any model beat the random-walk baseline with statistical significance? (2) Which model to deploy — cite MASE, DM p-value, EMH implications, and at least 3 non-accuracy criteria.	✓ Required
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4.2 Notebook Quality Standards

- All 23 questions answered — Q17 (MLP) and Q18 (LSTM) are required.
- All cells executed top-to-bottom — no skipped cells, no cells with `NameError` or `ImportError`.
- Every variable = `None` line replaced with your computed value — each is a named fill-in; leaving any as `None` will cause a runtime error.
- Every 🐞 **Reflect** cell contains your own reasoning with specific numbers — not a copy from the session notebook and not an AI-generated answer.
- All **SELF-CHECK** assert cells pass without error.
- Q4 stationarity table shows ADF and KPSS results for both raw prices and log-returns, with a conclusion referencing the EMH.
- Q9 Ljung-Box result states whether all $p > 0.05$ and what order adjustment was made if any failed.
- Q19 metrics table: all 7 metrics for all fitted models, sorted by MASE.
- Q22 Diebold-Mariano result states the p-value AND an explicit conclusion about statistical significance in the context of the COVID test period.
- Q23 investment memo cites actual MASE and DM p-value — no placeholder [brackets] remaining.

Do not submit: Any CSV files, the session solution notebook, Prophet HTML plots, TensorFlow SavedModel/ directories, `__pycache__`, or `.ipynb_checkpoints` folders. The only binary artefact beyond the notebook is `sp500_sarima_v1.pkl`.

5 COMMON MISTAKES TO AVOID

Patterns that produce wrong results or fail the submission check

✗ Fitting SARIMA without log-transforming first

The airline series has multiplicative seasonality — variance grows with the trend. ARIMA assumes constant-variance errors. Fitting on the raw scale violates this; the QQ-plot will show heavy tails and Ljung-Box will likely fail. Always fit on $\log(\text{train})$ and apply `np.exp()` to inverse-transform. If Q12's Ljung-Box fails, check log-transform first.

✗ Choosing ffill without justification in Q3

Q3 asks you to compare three strategies and explain which is best. Forward-fill repeats the last value — fine when the series is flat, but it creates a plateau during a rising trend. `interpolate(method='time')` is generally best for smooth seasonal series. Your Reflect answer must explain the choice, not just implement it.

X Using standard k-fold CV for the ML models

Standard sklearn k-fold shuffles the index, leaking future observations into training. This produces optimistically biased CV estimates. Always use TimeSeriesSplit (expanding window) for Q14. Walk-forward CV is not optional.

X Not using the history buffer in the recursive LightGBM forecast

Q15 requires a recursive loop: after predicting step i , append the prediction to history before computing lag features for step $i+1$. A common mistake is to read lag features from the original training data at every step — this means `lag_1` at step 13 equals `lag_1` at step 1, producing a periodic flat forecast. The recursive append is the whole point.

X Not discussing whether LSTM outperforms SARIMA given 360 training months

Q18 asks you to reflect on whether LSTM outperforms SARIMA with ~360 training points (vs the typical 96-point case). This is intentionally open-ended: with 360 months, LSTM may or may not win depending on the series. State your actual MASE comparison and explain the result. Do not simply copy the 'LSTM loses on small data' note — that was the 96-point airline case, not this 360-point S&P 500 case.

X Claiming 'MASE is better' without reporting the DM test

A lower MASE on 60 test observations does not prove genuine improvement, especially when the test period includes the COVID structural break. With high serial autocorrelation and regime change, effective sample size is much smaller than 60. Always report MASE and DM p-value together in Q22 and the memo.

X Empty or generic Reflect cells

Answers like 'the SARIMA model performed well' earn no marks. Reflect answers must cite specific numbers: which month had highest bias (Q20), at which horizon accuracy collapsed most (Q21), which lag feature was most important (Q14). Specificity distinguishes a data scientist who understands their model from one who ran the code and moved on.

Remember: The CIO's question is not 'which model has the lowest MASE?' — it is 'can we beat the random walk, and is that improvement statistically real?' MASE quantifies accuracy. The DM test quantifies whether the gap survives autocorrelation-corrected inference. Your Q23 memo must address both, plus the practical realities of regulatory explainability and structural break risk. A thoughtful memo that acknowledges EMH constraints and COVID limitations — even with MASE close to 1.0 — is more valuable than a metric table with a generic conclusion.